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WaveGuard: A flood monitoring and rescue system

Aritra Naha Likhan, Anirban Roy, Md. Shahidul Salim, Sk Imran Hossain *

Khulna University of Engineering & Technology, Khulna 9203, Bangladesh

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ABSTRACT

Floods hit millions of people every year around the world. In Bangladesh, flooding is a common phenomenon that is often caused by heavy rainfall, monsoon river flooding, and tropical cyclones. In Bangladesh, during the recent floods in 2024, more than 5 million people were affected due to inadequate early warning systems, coordination of rescue operations, and relief distribution. To address the requirements for sophisticated technologies and interconnected systems to monitor, manage, and contribute to flood-based disasters, we present WaveGuard. WaveGuard is a flood monitoring and rescue system web application embodying volunteer and victim authentication, dashboard for managing personal information, sharing nearby rescue locations, and effective collaboration between victims and volunteers, real-time location-based flood situation sharing, and providing alert insights about those locations with 77.57% accuracy in location recognition as a named entity and 94.41% accuracy in severity classification. A donation system, integrated via the bKash online payment interface, also aids fundraising. WaveGuard is developed using widely adopted technologies and frameworks, making it adaptable to other disaster response scenarios beyond floods.

Code metadata

Current code version	V1
Permanent link to code/repository used for this code version	https://github.com/ElsevierSoftwareX/SOFTX-D-25-00391
Permanent link to Reproducible Capsule	None
Legal Code License	MITLicense.
Code versioning system used	Git
Software code languages, tools, and services used	MERN, Python
Compilation requirements, operating environments & dependencies	Python 3.10, react-google-maps, axios, body-parser, concurrently, cors, dotenv, express, jsonwebtoken, leaflet, mongoose, morgan, multer, node-cron, nodemon, parser, puppeteer, socket.io, xlsx, jest-dom, react, user-event, chart.js, lucide-react, react-dom, react-redux, leaflet-routing-machine, react-router-dom, react-scripts, socket.io-client, web-vitals
If available Link to developer documentation/manual	https://github.com/AritraLikhan/WaveGuard/blob/main/README.md
Support email for questions	imran@cse.kuet.ac.bd

1. Motivation and significance

Flood management is a critical and complex challenge, particularly in flood-prone regions like Bangladesh, where floods annually submerge 20%–25% of the land area, and extreme events can affect up to 55%–60%, causing damages around US$1.0 billion [1]. The absence of robust systems for real-time monitoring, efficient rescue operations, and effective relief coordination exacerbates the impact. Key challenges include: (1) the lack of real-time data, hindering accurate flood prediction and response coordination; (2) communication breakdowns due to disrupted telecommunication and transport infrastructure; (3) inadequate rescue coordination, leaving many areas isolated; (4) chaotic resource allocation for essentials like food, medical supplies, and shelter; and (5) inefficient volunteer management, despite their critical role in rescue efforts.

Several IoT-based systems focused on water levels are primarily sensor-based and provide early warnings without incorporating user-generated data or real-time updates [2]. WaveGuard uses crowdsourced data and geospatial mapping, enabling real-time situation sharing.

* Corresponding author.
E-mail address: imran@cse.kuet.ac.bd (S.I. Hossain).

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Victim rescue solutions like drone-based rescue systems use object detection (e.g., YOLO) and provide GPS based emergency location information [3]. Another system offers live GPS tracking of rescue teams (L2S2), GNSS-based victim location capture [4]. WaveGuard ensures victim rescue by dynamically mapping rescue locations based on victims' continuous feedback through profile location update and chat system, integrating volunteer deployment based on proximity.

Multimodal LLM-based systems are proposed to enhance flood detection, communication, and preparation using Twitter data Named Entity Recognition (NER), computer vision on crowdsourced photos [5]. Another system applies Multimodal Neural Network combining text and hydrological data via Satellite Data (GSMaP), multilingual tweet classification, multilingual Word Embeddings via MUSE and fastText [6]. WaveGuard offers multilingual NLP system specifically designed for processing Bengali text with BERT-NER and Bengali Fasttext word embedding to achieve location-based crowdsourced flood severity data.

Despite advancements in flood monitoring and rescue systems, gaps remain in real-time data integration, communication, rescue coordination, and volunteer management. Many solutions primarily rely on IoT sensors or satellite data, lacking integration of user-generated data or effective communication tools [2]. Additionally, existing systems often do not support multilingual crowdsourced data or provide real-time flood situation updates [4]. WaveGuard addresses these gaps by integrating NLP, geospatial mapping, and crowdsourcing into a unified platform. It enables real-time flood situation sharing, volunteer–victim authentication, and better coordination through a chat system, offering more interactive and actionable communication compared to sensor-only platforms [3]. WaveGuard also features NLP-based severity detection in Bengali, enhancing crowdsourced reporting in local contexts, which is absent in many solutions [6]. Moreover, its donation system via bKash ensures transparency in fundraising for verified NGOs, further setting it apart from existing platforms.

WaveGuard, built on scalable technologies like the MERN stack and Python, addresses flood challenges and can be adapted to other disasters like earthquakes, hurricanes, and wildfires by effective volunteer–victim coordination, it is crucial for rapid assistance and long-term recovery by promoting community resilience and psychological support [7].

2. Software description

A structural description of this website is given in Section 2.1. Subsequently, details on the implementation of each feature are provided in Section 2.2.

2.1. Software architecture

The website is built on the MERN stack, a combination of four components : MongoDB, React, Express.js, Node.js.[Fig. 1]

1. **MongoDB**: A NoSQL database for storage and retrieval of semi-structured data in JSON format.
2. **Express.js**: A lightweight Node.js framework for building RESTful APIs.
3. **React**: A responsive and interactive frontend framework for a seamless user experience.
4. **Node.js**: A server-side JavaScript runtime environment that handles concurrent requests properly.

Apart from these, a Python script, integrated via a Flask API, has also been used to utilize NLP models for generating alert data. The Node.js (Express) backend communicates with the Python Flask backend by sending HTTP requests; after scraping posts, Node.js uses Axios library to send a POST request to the Flask API endpoint with the data. Flask processes the data using NER and embedding models and sends back a response. This way, even though Node.js uses JavaScript and Flask uses Python, they communicate through standard HTTP requests and responses (REST API).

The website consists of five main features: volunteer and victim registration, user authentication, dashboard for volunteers and victims, donation and aid management, crowdsourced report-based alert. Each feature is described in Section 2.2.

2.2. Software functionalities

2.2.1. Volunteer and victim registration

In WaveGuard, users register by providing details like name, contact, email, password, location, and selecting a role (volunteer or victim). The backend validates the input, saves the information to the database, and sends a verification email. Once verified, the user can log in. The registration process securely handles user data submission for account creation.

```
# Pseudocode of registration backend logic
Function registerUser(request):
    Extract email, password, role, location
    Generate emailVerificationToken
    Create user with isVerified = false
    Save user to database
    Send verification email with token
    Return success response
```

2.2.2. User authentication

WaveGuard offers secure authentication using Role-Based Access Control (RBAC). The frontend collects the user's email and password, sends a POST request to the backend, which verifies the credentials and returns a JWT token along with the user's profile from MongoDB. The token is stored in local storage or cookies for future requests. The authMiddleware restricts access to sensitive data like location information. Upon successful authentication, users are redirected to the appropriate dashboard based on their role (volunteer or victim).

```
# Pseudocode of authentication backend logic
Function login(request):
    Get email and password
    Find user by email
    If not found or email not verified:
        Return error
    If password matches:
        Generate JWT token
        Return user data + token
    Else:
        Return invalid credentials
```

2.2.3. Dashboard for volunteers and victims

The WaveGuard dashboard streamlines volunteer efforts and supports victims during flood emergencies, combining front-end and back-end functionalities. Built with React.js, it includes components for profile management, shelter location via Google Maps, real-time messaging, and notifications. The backend, powered by Node.js and Express.js, handles user operations, authentication, and data management. Volunteers and victims can access profiles, view nearby shelters, and receive directions for optimized rescue efforts. The integrated chat system and geospatial data improve coordination, communication, and resource allocation in flood response [Fig. 2].

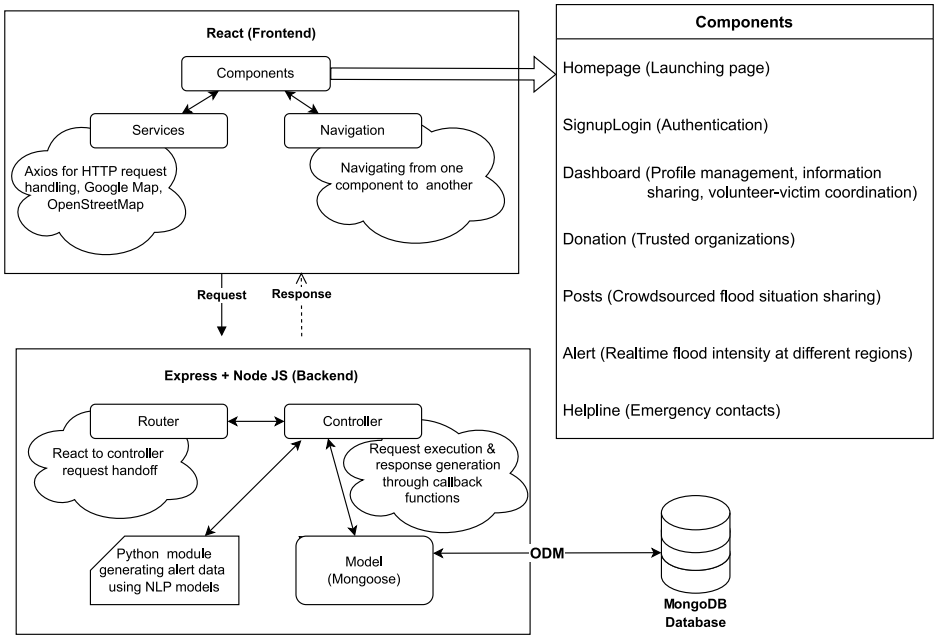


Fig. 1. WaveGuard architecture.

```
# Pseudocode of dashboard backend logic
Function updateProfile(userId, newData):
    Find user by ID
    Update fields using newData
    Save user
    Return updated user info

Function getNearbyUsers(role, locationQuery):
    Parse query terms
    Create regex from location terms
    Find users by:
        role == targetRole
        location matches regex
    Return list of users

Function getNearbyShelters(location):
    Split into zilla and upazila
    Find shelter document by zilla
    Find shelters in given upazila
    Return matched shelters

Function saveMessage(request):
    Extract senderId,receiverId,message from request
    Create chatMessage with senderId, receiverId
    Save chatMessage
    Return chatMessage
```

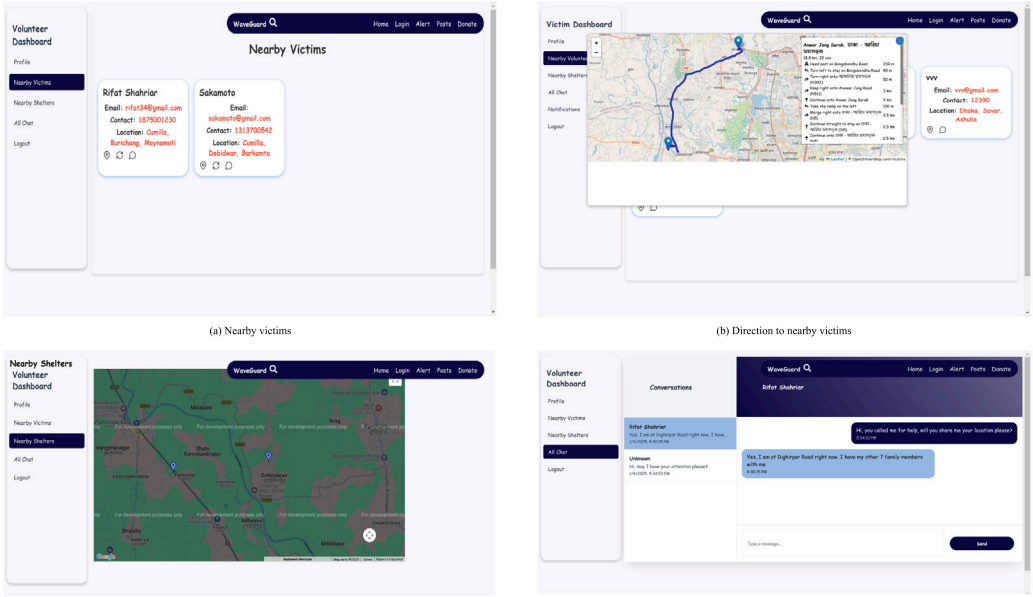
2.2.4. Donation and aid management

WaveGuard enables donation collection and distribution through bKash, ensuring transparency in resource allocation [Fig. 3]. The donation campaigns and organizations are displayed via a 'DonationCard' component. Selection of a specific organization redirects to the bKash interface, providing an interface to fill in donor's name, email and donation amount into a form and perform payment through subsequent gateway steps.

```
# Pseudocode of donation backend logic
Function saveDonation(request):
    Extract tran_id, name, amount, orgId
    Generate ID if missing
    Save donation to database
    Return success
Function executeDonation(request):
    Navigate to org-specific donation URL
    Fill name, email, amount
    Complete gateway steps
    Return success/failure
```

2.2.5. Crowdsourced report-based alert

The feature allows users to post updates, analyzes content to extract location and severity, and visualizes data for generating warnings or insights [Fig. 4]. Posts include features like upvoting and downvoting to prioritize alerts. Incident severity is visualized using a bar chart, with height and color indicating severity levels. On the backend, NLP techniques — BERT Multilingual Cased NER model for Named Entity Recognition and Bengali FastText embeddings for semantic similarity analysis — are used to extract locations and classify severity. The NER model performs approximately with 77.57% accuracy [8] in location recognition as a named entity, while the embedding model demonstrates approximately 94.41% accuracy in severity classification [9]. The processed data is saved to an Excel file for generating warnings or further analysis.



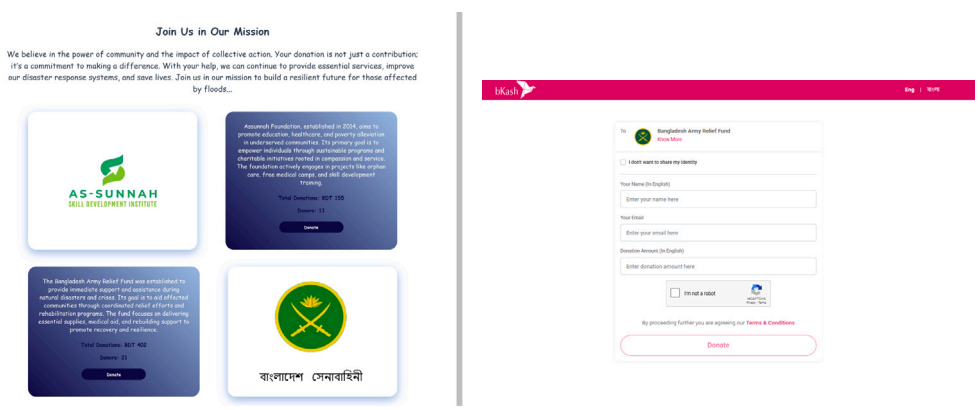
(a) Nearby victims

(b) Direction to nearby victims

(c) Nearby shelters

(d) Chat interface

Fig. 2. Dashboard sections (a) Nearby victims: detailed contact information, chat interface and direction-on-map shortcuts of victims close to a volunteer; (b) Direction to nearby victims: route from a volunteer to a victim on map using OpenStreetMap; (c) Nearby shelters: shelter close to victim's location marked on map using Google Maps API; (d) Chat interface: an interface for volunteers and victims to send and receive text messages.



(a) Fundraising organizations

(b) bKash interface

Fig. 3. Donation sections (a) Fundraising organizations: demonstrating a list of organizations that receive donations through bKash; (b) bKash interface: providing an interface to fill in donor's name, email and donation amount into a form and payment through subsequent gateway steps (not displayed due to user privacy) to make a donation to the selected organization.

```
# Pseudocode of alert data backend logic
Function startScraping():
    Launch Puppeteer
    Scroll & scrape up to 10 posts
    Send posts to Flask API
Flask Function processPosts(posts):
    For each post:
        Run NER to extract location
        For each word in post:
            Compute similarity with severity vectors
            If similarity > threshold:
                Increment severity count for location
    Save location + severity data to Excel
```

3. Illustrative examples

To better demonstrate the major functions of the software, an explanatory video has been provided as a supplementary video file. The video illustrates the key workflows and features described in the article.

4. Impact

WaveGuard addresses a critical challenge in disaster management: effective co-ordination between volunteers and victims. By integrating a real-time chat system and precise location sharing (both textual and visual), the software streamlines communication, ensuring rapid and targeted responses. This reduces duplication of efforts and helps deliver assistance where it is most urgently needed.

A key innovation is WaveGuard's use of geospatial data and real-time reporting to optimize resource allocation. Volunteers can be directed to the most critical areas, and shelters can be prepared in

(a) Public posts

(b) Location-based severity alert

Fig. 4. (a) Public posts: text and image based posts uploaded by website users; (b) Location-based severity alert: visualizing near-realtime flood severity at districts extracting location and severity data from public text posts using NLP techniques.

advance based on predictive insights. These capabilities significantly improve the efficiency of disaster response compared to traditional manual coordination.

Furthermore, WaveGuard incorporates natural language processing (NLP) to analyze social media reports and other textual data, enabling severity estimation and area-specific visualizations of flood impacts. Unlike top-down government warnings—which may lack local accuracy — WaveGuard collects and interprets data at the grassroots level, enhancing the timeliness and reliability of alerts. This has the potential to reduce casualties and economic loss effectively supported by AI simulated data.

In the context of Bangladesh, where 20%–25% of the land area is flooded annually, and extreme floods affect up to 60% of the country, the economic and humanitarian impact is profound. WaveGuard presents a scalable and cost-effective solution to help mitigate this recurring crisis, supporting national-level flood preparedness and management efforts.

WaveGuard also opens up new research directions, such as localized flood prediction models using social media inputs, and improvements in named entity recognition (NER) for geographic terms in low-resource languages. Currently, the NER model supports subdistrict level recognition within Bangladesh. Since the current NLP accuracy of interpreting social media reports depends on the accuracy of pre-trained models, expanding this to finer-grained geographic area recognition and severity classification through additional training data from real-world scenarios could enable even more precise situational awareness.

The ethical issues of data privacy such as exposure of sensitive personal or geolocation data, commercial exploitation of user data require further attention which have not been addressed by WaveGuard.

Additionally, although shelter locator features have been integrated using static maps, displaying real-time flood overlays on dynamic maps remains a challenge due to limitations of publicly available mapping services. This represents a promising area for further development in large-scale deployments. Although WaveGuard incorporates features targeting maximum success, actual outcome remains a challenge since no pilot test was conducted.

5. Conclusions

Floods represent a growing threat to millions of people, especially in flood-prone countries like Bangladesh. While existing systems provide some relief, many areas still lack timely intervention and efficient coordination. WaveGuard seeks to address these challenges by integrating advanced technologies such as AI, NLP, geospatial mapping, and crowdsourcing into a unified platform. By streamlining flood monitoring, rescue coordination, and relief efforts, WaveGuard can significantly improve the efficiency and effectiveness of disaster response, ultimately saving lives and reducing the impact of floods on communities. The project leverages common and scalable technologies, which facilitates adaptation and deployment in other disaster response contexts, thereby broadening its impact potential.

CRediT authorship contribution statement

Aritra Naha Likhan: Writing – original draft, Software, Conceptualization. **Anirban Roy:** Writing – original draft, Software, Data curation. **Md. Shahidul Salim:** Supervision, Resources, Methodology. **Sk Imran Hossain:** Writing – review & editing, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the manuscript. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.softx.2025.102361.

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